

PumpRupee

Pricing pump energy waste in rupees

A clamp-on audit kit that tells you if a VFD will pay back for THAT pump.

No flow meter

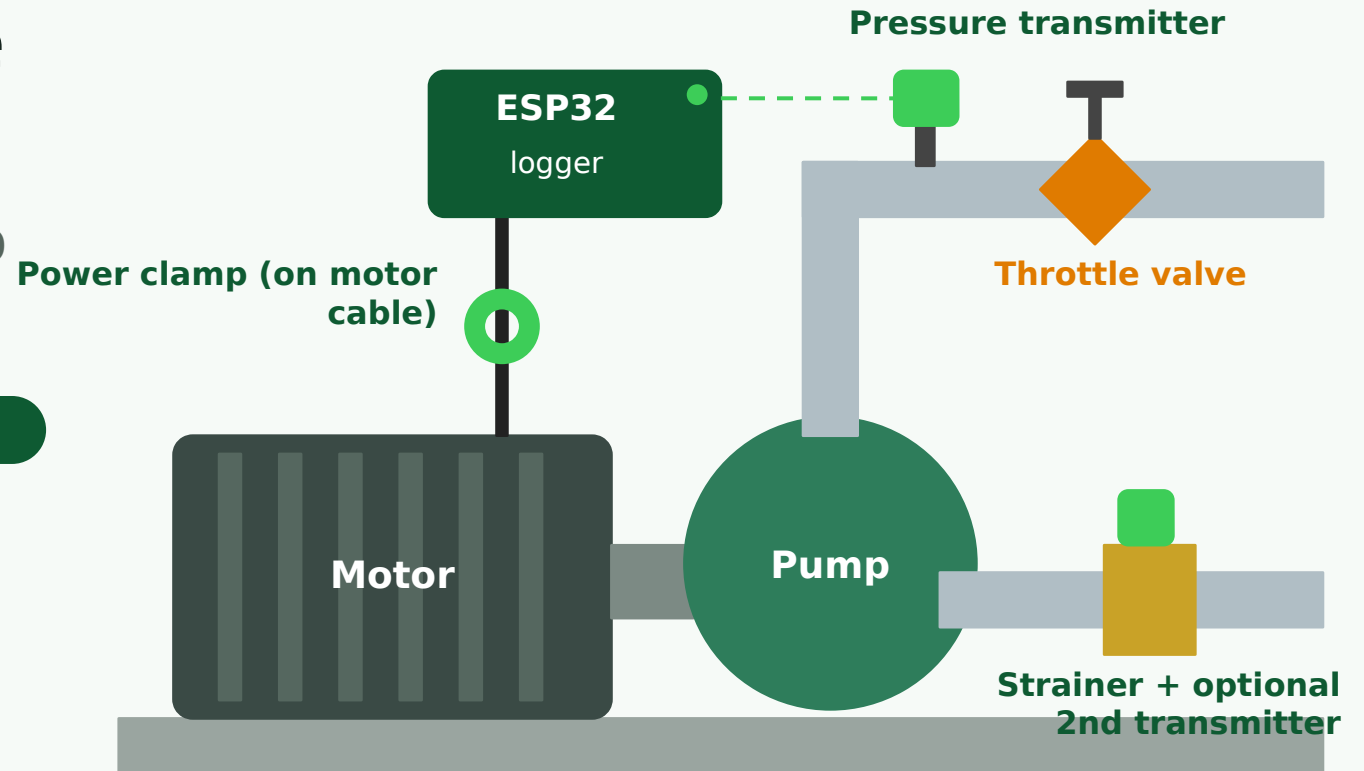
No shutdown

₹ waste by cause

Schneider Electric Yuva Yodha Energy Tech Hackathon 2026

Track: Smart Manufacturing
Electrify · Automate · Digitalise

Team ANS_4X



All results are simulated (Grundfos NB 65-160/157, 11 kW). Prices are indicative online prices, not quotes.

Problem: Invisible Waste

Oversized pumps run throttled. The extra energy is burnt across a valve, every hour.



The owner is blind to it.

Flow is hard and costly to measure. An energy-audit practitioner reports that 30-40% of ultrasonic flow readings go wrong. Head (pressure) and electrical power are easy to measure.

Most common cause

Wrong sizing leads to throttling
(practitioner report)

83-93%

of simulated waste is the valve (72-87% under stress)

8-45%

VFD saving across simulated cases; 5-40% reported in practice

Simulated figures: Grundfos NB 65-160/157, 11 kW. The 83-93% valve share is measured against a speed-matched, healthy, clean ideal.

Why Today's Options Rarely Reach MSMEs

Each option is valuable. None is a low-cost entry point that ends in a ₹ decision.

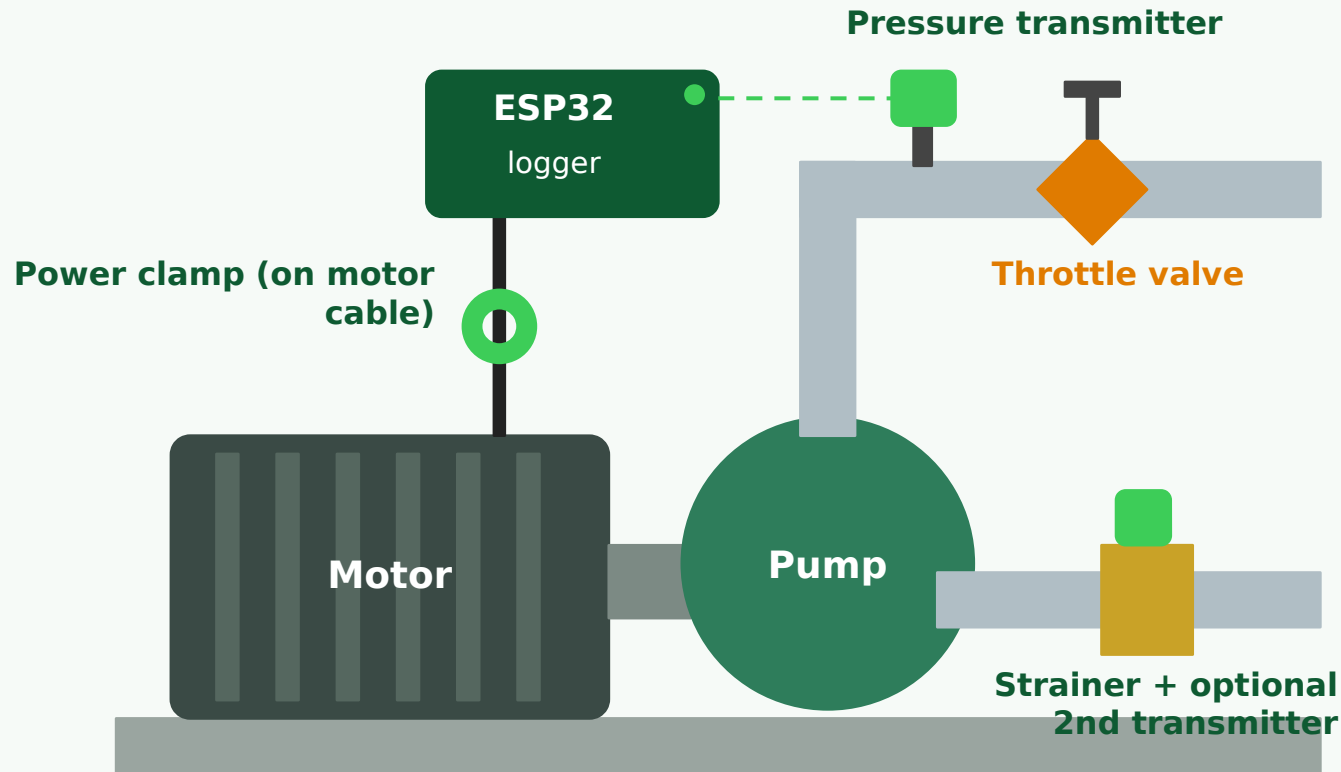
	Flow-meter retrofit	One-time manual audit	Full monitoring platform	PumpRupee
Upfront cost, small plant	Meter + install	Fee per visit	Typically higher	₹6,100-13,300
No plant shutdown	—	✓	—	✓
Answer in ₹	✗	✓	—	✓
Waste split by cause	✗	✓	—	✓
Pump-specific VFD verdict	✗	✓	—	✓
Verifies saving afterwards	—	—	✓	✓
Suited to small MSMEs	—	✗	—	✓

✓ yes ~ partly / varies by product ✗ no. Our general assessment, not a product test. PumpRupee's ticks are design targets (simulated, not field-proven).

Full platforms are the natural NEXT step after PumpRupee, not a rival.

Our Solution: A ₹ Audit Kit

A clamp-on kit turns power + pressure readings into “₹ wasted per year, and why”.



VFD = control speed instead of strangling the pipe.



₹ waste per year

Pump energy waste priced in rupees



Waste by cause

Valve, impeller wear, clogged strainer



Cheapest fix first + setpoint

Lowest safe VFD setpoint



VFD payback verdict

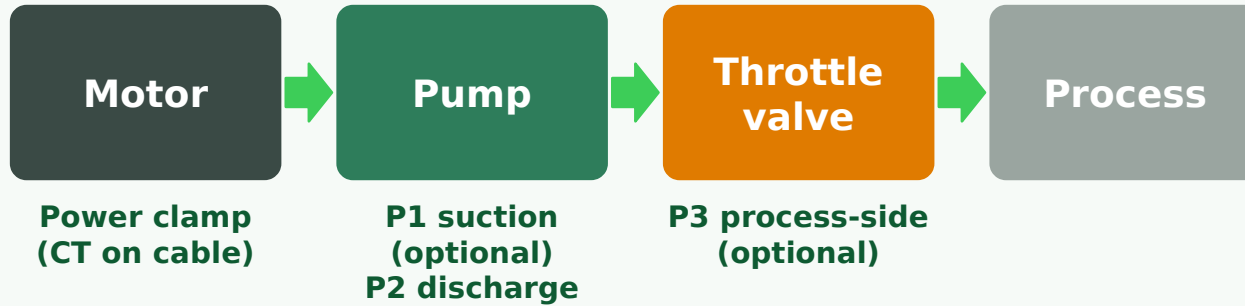
For THIS pump, before capex

No flow meter · no pipe cutting · no shutdown
Sensors on existing gauge ports; owner keeps the data.

Audits and recommends. Does not control the pump.

How It Works: Install & Pump Curve

Clamp on, log for a week or two, and let the maths find the waste. Curves below are the real pump.



Minimum kit: power clamp + discharge pressure (P2).

Add P3 (process-side) when a VFD purchase is on the table: it measures the system curve and reveals a clogged strainer.

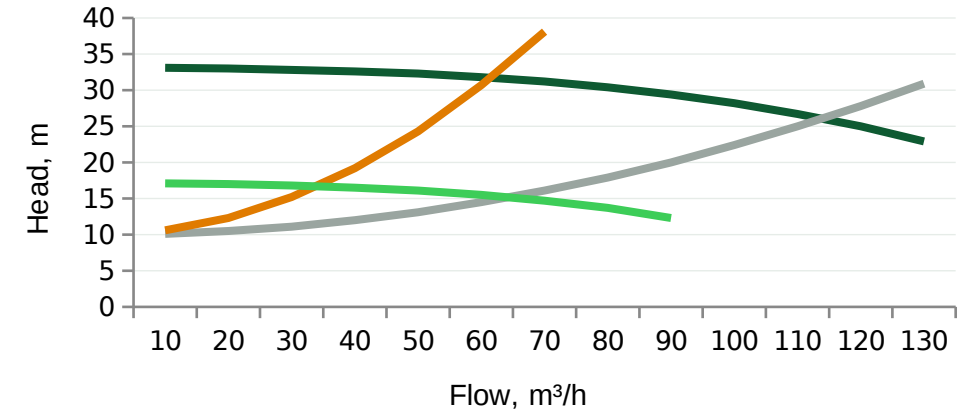
P2 + power Waste priced in ₹ (basic)

+ P3 (about ₹3,068) Error 10.5 → 1.4 pts (stress, simulated)

+ P1 / strainer ΔP Clog made visible

System curve = what the pipe network demands. Pump curve = what the pump can deliver at a given speed.

Grundfos NB 65-160/157: head vs flow



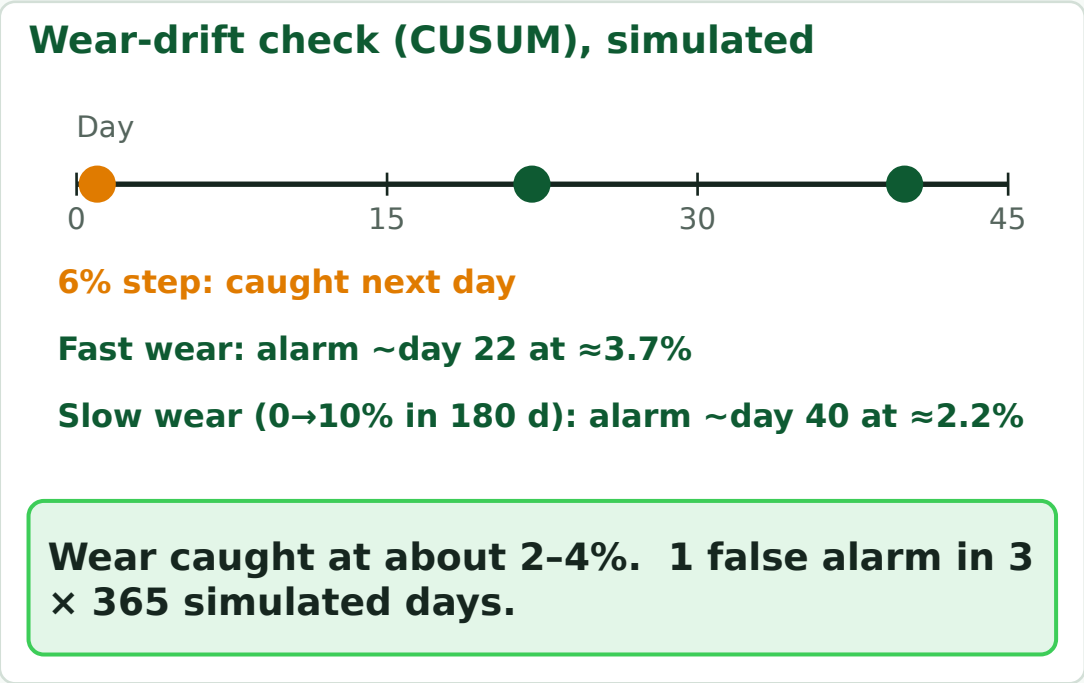
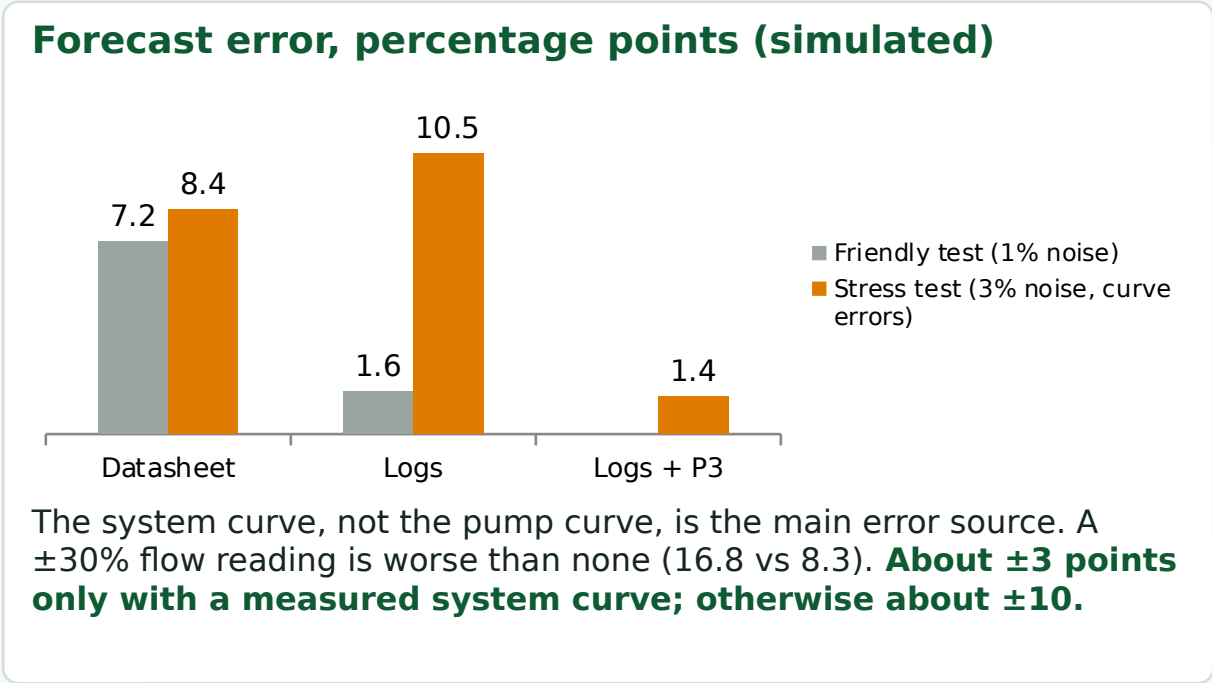
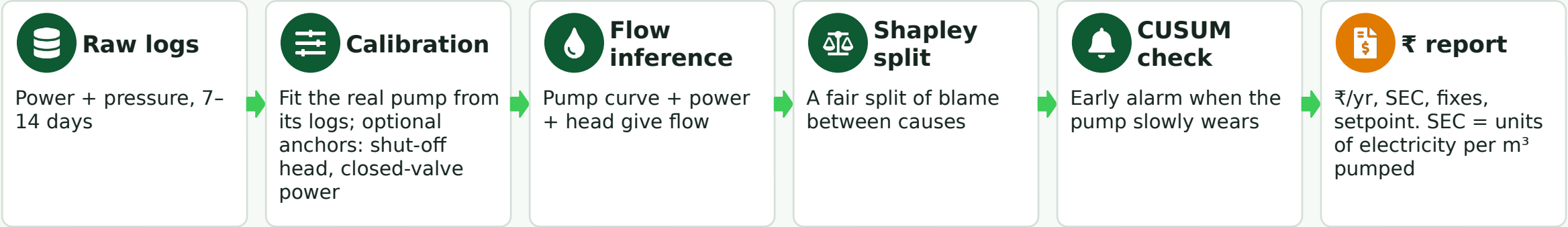
- Pump, full speed (fit to maker data)
- System, valve open (assumed 10 m lift)
- System, throttled
- Pump slowed by VFD (72% speed)

Off-peak hour (60% of peak demand, 61.6 m³/h)

Throttled: the pump delivers 31.7 m, the pipe needs 15.4 m (incl. 5% margin). The valve burns the gap. VFD: slow to 72% speed. That gap is what PumpRuppee prices in ₹.

Analytics & Honest Accuracy

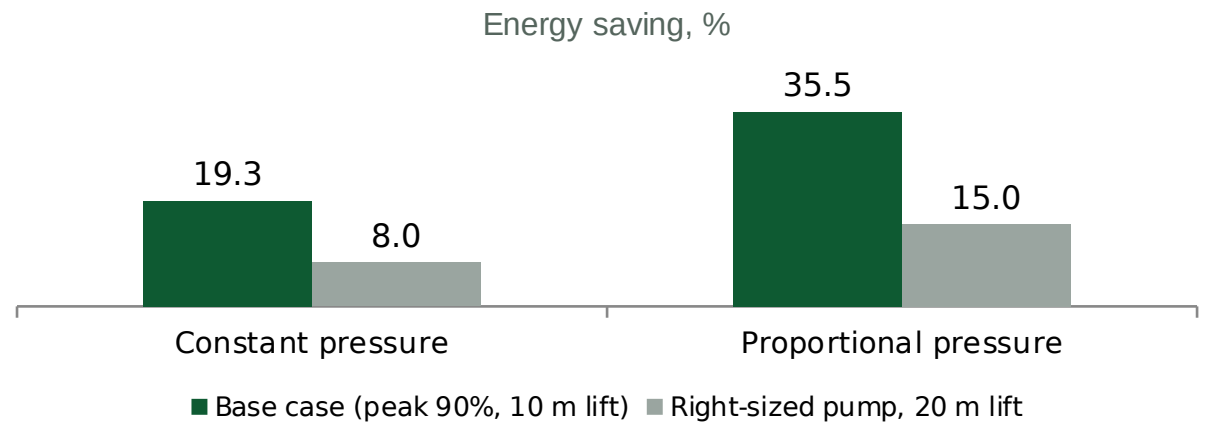
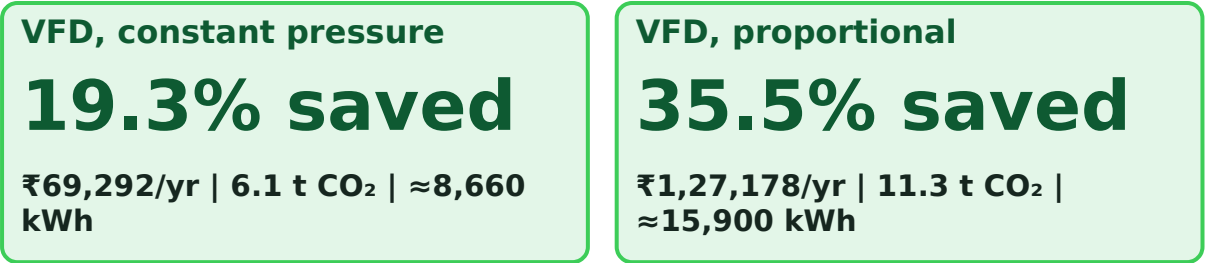
Six plain steps from raw logs to a ₹ report, with the uncertainty stated.



Results (Simulated)

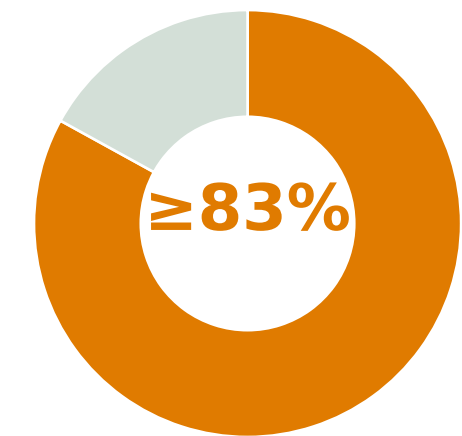
SIMULATED - Grundfos NB 65-160/157, 11 kW, IE3

Assumed: 10 m static lift | 5% head margin | 16 h/day | 300 days/yr | ₹8/kWh | VFD efficiency 0.97 | 0.710 kg CO₂/kWh. Peak demand 90% of rated flow.



Range: 8-33% (constant), 15-45% (proportional). 54-60% for heavily oversized pumps is an upper bound only. ₹, kWh and CO₂ are consistent: kWh = ₹ ÷ 8; CO₂ = kWh × 0.710 (arithmetic).

Where the waste goes



Valve share 83-93%
(72-87% under stress). Donut shows the low end. Measured against a speed-matched, healthy, clean ideal.

Clog blind spot
A clogged strainer is invisible with power + pressure alone. One extra pressure sensor recovers about ₹6,000-6,600/yr of misattributed waste.

Cost & VFD Verdict

INDICATIVE ONLINE PRICES, NOT QUOTES

Bill of materials (preliminary)

ESP32 dev board ₹399-449

Power clamp, SCT-013 100 A ₹349 + GST to ₹729

Voltage sensor ZMPT101B ₹92-289

Pressure transmitter, basic ₹2,100-3,068

Enclosure, supply, 4-20 mA, wiring ₹2,000 (assumed)

Basic single-phase ≈ ₹6,100

3-phase + 2nd transmitter ≈ ₹10,200

3-phase + process-side (P3) ≈ ₹13,300

With ₹12,000 transmitters ≈ ₹40,100

Hobby-grade clamps; not a certified meter. Volume cost: not estimated (no quotes).

Value per pump

Cost per pump audited

Kit ÷ N pumps. N = 10: ₹610-1,330

Audit time per pump

Install (assumed ½ day) + 7-14 days logging + report

Kit throughput

About 25-50 pumps/yr per kit at 7-14 days each (arithmetic, excludes travel)

Kit payback

₹10,200 ≈ 1.8 months of the base-case saving of ₹69,292/yr, if the audit leads to that VFD (illustrative)

VFD verdict (simulated)

Oversized pump

4-19 months

GO: buy the VFD

Right-sized, high lift

2-4+ years

WAIT: fix cheaper things first

VFD ₹29,000-94,000 + about ₹15,000 install (assumed): ₹50,200-1,09,400 in total.

A ~₹10k kit can stop a wrong ₹50k-1.1 lakh VFD buy, or unlock ₹69,292-1,27,178/yr (simulated)

Innovation: What's New, and What Isn't

We build on proven tools. The new part is the integration for MSMEs.

Already exists (we build on it)

- **KSB PumpMeter** pump monitoring
- **US DOE PSAT** pump-system assessment tool
- **Samotics** motor-current pump efficiency
- **Schneider Altivar Process** sensorless flow in the drive
- **EcoStruxure, Indian IoT vendors, ESCOs** energy monitoring and audits
- **Shapley split, CUSUM, calibration** known methods, applied here



What PumpRupee adds: five answers

- ✓ **How many ₹ is this pump wasting, and why?**
waste priced by cause
- ✓ **Is an extra sensor worth buying?**
sensor value priced in ₹
- ✓ **Will a VFD pay back for THIS pump?**
verdict before capex
- ✓ **Did the fix work?**
built-in before/after check
- ✓ **Can an MSME afford it?**
₹6,100–13,300, no flow meter

“What's new?” Most tools tell you how a pump is running. PumpRupee tells an MSME owner what to buy, in rupees, before spending, and proves it afterwards.

We found no low-cost shared kit that combines these. That is a search result, not proof that none exists.

Ecosystem Fit: Audit → Drive → Verify

PumpRuppee is the low-cost front door to VFD and energy-management adoption.



Electrify

Makes motor electrical energy visible with a clamp-on power measurement and points owners to efficient drives.



Automate

Shows which pumps justify a drive and the lowest safe setpoint. The drive does the controlling; PumpRuppee never does.



Digitalise

Logs become a ₹ business case and a before/after proof. The owner keeps the data and can use any drive.

1

PumpRuppee audit

Clamp-on, 7-14 days

2

₹ business case

Waste by cause + VFD verdict

3

VFD installed

e.g. Altivar Process

4

PumpRuppee re-measures

Same kit, after the fix

5

Verified saving

Feeds energy management, e.g. EcoStruxure

Why this helps drive and EMS adoption

MSMEs that skip audits today get a ₹ case they understand, a pump-specific verdict before capex, and proof after.

Smart Manufacturing fit

Energy efficiency · productivity · digitalisation · sustainability

Concept only: no drive or platform integration built.

Impact, Scalability & Roadmap

IMPACT = SIMULATED, per 11 kW pump

₹69k-1.27 lakh
per year

≈8,660-15,900
kWh per year

6.1-11.3
t CO₂ per year

Scale-up illustration, not a projection: N pumps × per-pump saving. N = 10: ₹6,92,920-12,71,780/yr and 61-113 t CO₂/yr.

Scalability

Single pump

Plant (shared kit)

Cluster / ESCO / channel partner

Bundled with VFD & EMS offers

For: MSME factories, water utilities, textile, food processing, HVAC chilled water.

Roadmap

1. Simulation
2. Bench test, real pump + sensors
3. Pilot, 1-2 plants, measured system curve
4. Validate forecast error
5. Multi-pump / multi-plant rollout

DONE

NEXT

PLANNED

PLANNED

PLANNED

Honest limits. Actually tested: nothing yet.

- Simulation only: no lab/field data, no real quotes
- System-curve fit is optimistic (same form as the simulated plant)
- Single pump, steady state; no parallel pumps, closed loops, drives, NPSH or control dynamics
- Valve share partly by construction; 54-60% is an upper bound

Team ANS_4X

Schneider Electric Yuva Yodha 2026 | Smart Manufacturing | [Institution]



[Name]
[Role]
[Skill: one line relevant to pumps, sensors, data or energy]



[Name]
[Role]
[Skill: one line relevant to pumps, sensors, data or energy]



[Name]
[Role]
[Skill: one line relevant to pumps, sensors, data or energy]



[Name]
[Role]
[Skill: one line relevant to pumps, sensors, data or energy]

Cheap kit. Clever inference. Answers in rupees. Honest about limits.

Replace the photo circles and [placeholders]; delete unused cards (1-4 members).